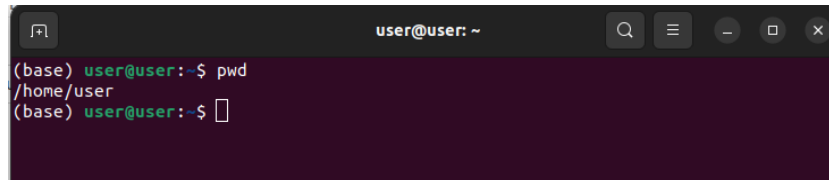


CS110 – Computing Lab

Week – 1 (11th Jan to 18th Jan 2024)

Try the following linux commands.

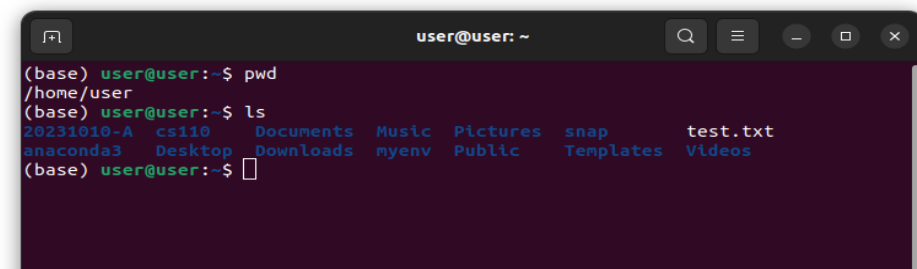
1. After login, open the **terminal** application and run the command *pwd*, and check if you can see a screen like below. (Shortcut key: ctrl+alt+T)



```
(base) user@user: ~  
(base) user@user:~$ pwd  
/home/user  
(base) user@user:~$
```

The *pwd* (**present working directory**) command shows the directory that you are currently working on. It shows “/home/user”. It means, you are presently under the “user” directory, which is under another directory name “home”. It is a typical location that you will be placed after logging in. In the above example, “user” is the name of my login. If you have login with some other name, in the place of “user”, you will see your login name. The */home/user/* is your work space. Anything you do (like creating directory or file will be created under this work space.

2. Type the command *ls*. It will show the list of files and directories created under */home/user/*. You will see a screen like below.



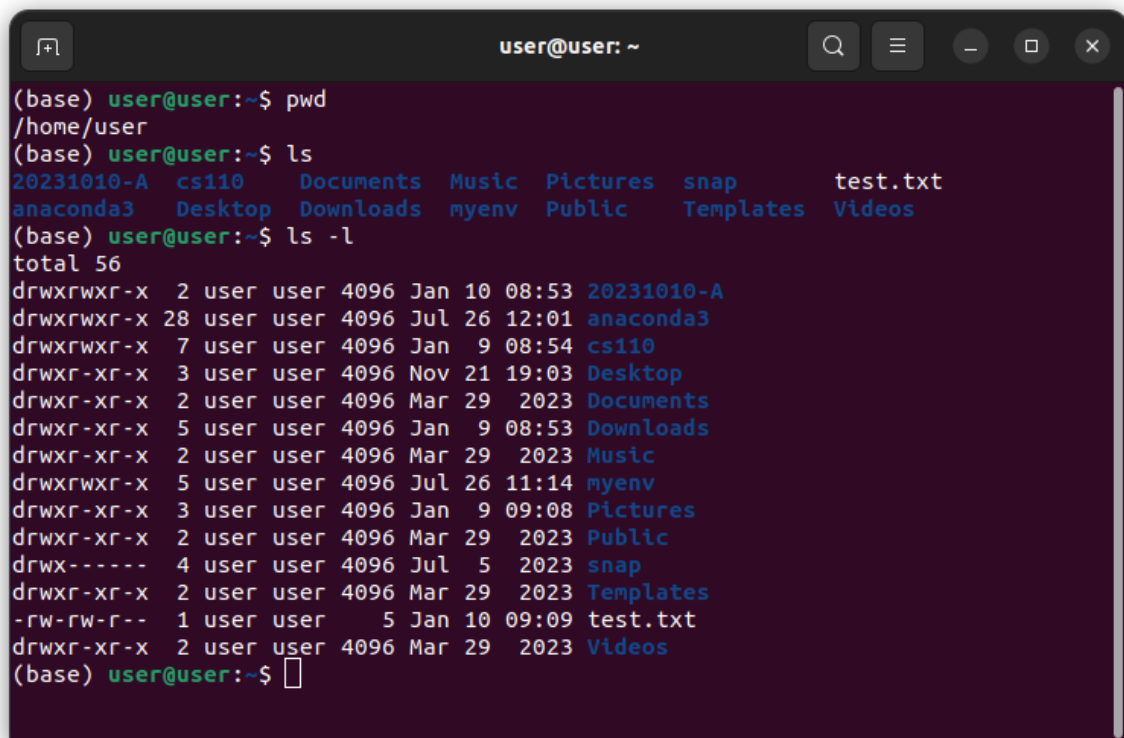
```
(base) user@user: ~  
/home/user  
(base) user@user:~$ ls  
20231010-A  cs110  Documents  Music  Pictures  snap      test.txt  
anaconda3  Desktop Downloads  myenv   Public   Templates Videos  
(base) user@user:~$
```

The names in **blue** color are the directories and the names in **white** are the files. Depending on your system configuration, you may see with different colors.

You can run the ls command with various options. The set of options that you can use are listed below.

Options	Description
-l	known as a long format that displays detailed information about files and directories.
-a	Represent all files Include hidden files and directories in the listing.
-t	Sort files and directories by their last modification time, displaying the most recently modified ones first.
-r	known as reverse order which is used to reverse the default order of listing.
-S	Sort files and directories by their sizes, listing the largest ones first.
-R	List files and directories recursively, including subdirectories.
-i	known as inode which displays the index number (inode) of each file and directory.
-g	known as group which displays the group ownership of files and directories instead of the owner.
-h	Print file sizes in human-readable format (e.g., 1K, 234M, 2G).
-d	List directories themselves, rather than their contents.

If your run with -l option you will see a screen like below.

A terminal window titled 'user@user: ~' with standard window controls. The user has entered a series of commands: 'pwd' (returns '/home/user'), 'ls' (returns a simple listing of directories and files), and 'ls -l' (returns a detailed long-format listing). The 'ls -l' output shows permissions, owner, group, size, and timestamps for each file and directory. The files listed are '20231010-A', 'anaconda3', 'cs110', 'Desktop', 'Downloads', 'Music', 'Pictures', 'Public', 'snap', 'Templates', 'test.txt', and 'Videos'.

```
(base) user@user:~$ pwd
/home/user
(base) user@user:~$ ls
20231010-A  cs110    Documents  Music  Pictures  snap    test.txt
anaconda3  Desktop  Downloads  myenv  Public    Templates Videos
(base) user@user:~$ ls -l
total 56
drwxrwxr-x  2 user user 4096 Jan 10 08:53 20231010-A
drwxrwxr-x 28 user user 4096 Jul 26 12:01 anaconda3
drwxrwxr-x  7 user user 4096 Jan  9 08:54 cs110
drwxr-xr-x  3 user user 4096 Nov 21 19:03 Desktop
drwxr-xr-x  2 user user 4096 Mar 29 2023 Documents
drwxr-xr-x  5 user user 4096 Jan  9 08:53 Downloads
drwxr-xr-x  2 user user 4096 Mar 29 2023 Music
drwxrwxr-x  5 user user 4096 Jul 26 11:14 myenv
drwxr-xr-x  3 user user 4096 Jan  9 09:08 Pictures
drwxr-xr-x  2 user user 4096 Mar 29 2023 Public
drwx----- 4 user user 4096 Jul  5 2023 snap
drwxr-xr-x  2 user user 4096 Mar 29 2023 Templates
-rw-rw-r--  1 user user   5 Jan 10 09:09 test.txt
drwxr-xr-x  2 user user 4096 Mar 29 2023 Videos
(base) user@user:~$
```

It will show the files and directories with their detailed information. The lines beginning with **d** indicate directories, and the lines beginning with hyphen (–) indicate normal files. Every line has 7 fields, as explained below.

The following descriptions consider the first line ***“drwxrwxr-x 2 user user 4096 Jan 10 08:53 20231010-A”***

Field 1 – File Permissions: The first field defines the type and permissions

Ex. The first field ***drwxrwxr-x***

The first character defines if it is a file or directory. Following are the possible file type options.

- – normal file
- d : directory
- s : socket file
- l : link file

Permissions: The next six(06) characters ***rw-rw-r-x*** define the permission. Every 3 characters specify read, write, execute permissions for user, group and others respectively in order. User is your user name, group is the name of the group your user name belong to, others is the other users belonging to the same group you belong too. Taking the above example, ***rw-rw-r-x*** indicates read-write-execute permission for user name ***user***, read-write-execute permission for group name ***user***, and read-execute permission for others respectively. For details on user-group-others and permission, you can refer to <https://www.redhat.com/sysadmin/manage-permissions>.

Field 2 – Number of links: Second field specifies the number of links for that directory. In this example, 2 indicates only one link to this directory.

Field 3 – Owner: Third field specifies owner of the directory. In this example, this directory is owned by username ‘user’.

Field 4 – Group: Fourth field specifies the group of the directory. In this example, this directory belongs to “user” group.

Field 5 – Size: Fifth field specifies the size of directory in bytes. In this example, ‘4096’ indicates the directory size in bytes.

Field 6 – Last modified date and time: Sixth field specifies the date and time of the last modification of the directory. In this example, ‘***Jan 10 08:53***’ specifies the last modification time of the directory.

Field 7 – File name: The last field is the name of the directory. In this example, the directory name is ***20231010-A***.

3. Create two directories with the names <rollno>-A and <rollno>-B in your home directory.

[Command for making directory: *mkdir directory_name*]

Ex. If your roll number is 20231010, then the names of the directories should be 20231010-A and 20231010-B

4. Now change your current directory to one of the directory you have just created.

[Command for changing current directory: *cd directory_name*]

5. Open **gedit** text editor, and type the following C program

[Command to open gedit text editor: *gedit*]

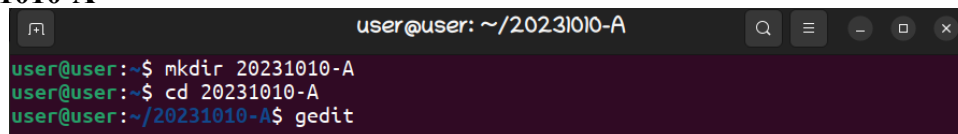
```
#include<stdio.h>

int main(){

printf("Hello World\n");
return 0;

}
```

Save the file with the above program with the file name "helloworld.c" in the directory **20231010-A**

A terminal window with a dark background. The title bar shows 'user@user: ~/20231010-A'. The terminal content shows three commands and their outputs: 'mkdir 20231010-A', 'cd 20231010-A', and 'gedit'.

```
user@user: ~/20231010-A
user@user:~$ mkdir 20231010-A
user@user:~$ cd 20231010-A
user@user:~/20231010-A$ gedit
```

6. Now, you've created the directories and the C program in the specified directory. Compile and run the program with the following commands:

1. # Compile the C program

[command for compiling *gcc code_file_name.c*]

Ex. *gcc helloworld.c*

- Once the program gets compiled successfully a compiled file (a.out) will be generated. You can see this file by entering **ls** command to the terminal.

2. # Run the compiled program

./a.out

3. This will print "Hello World" to the terminal.

4. # Compile the C program with custom compiled file name.

[command for compiling *gcc code_file_name.c -o custom_name*]

Ex. *gcc helloworld.c -o helloworld*

- Once the program gets compiled successfully a compiled file (helloworld) will be generated. You can see this file by entering **ls** command to the terminal.

5. # Run the compiled program
./ helloworld
6. This will print "Hello World" to the terminal.



```
user@user: ~/20231010-A
user@user:~/20231010-A$ ls
helloworld.c
user@user:~/20231010-A$ gcc helloworld.c
user@user:~/20231010-A$ ls
a.out  helloworld.c
user@user:~/20231010-A$ ./a.out
Hello World
user@user:~/20231010-A$ gcc helloworld.c -o helloworld
user@user:~/20231010-A$ ls
a.out  helloworld  helloworld.c
user@user:~/20231010-A$ ./helloworld
Hello World
user@user:~/20231010-A$
```